STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 4: Laws of Motion

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. What measures translational inertia?
2. Second law at constant mass?
3. Do action-reaction forces act on one body?
4. Momentum formula?
5. Net force at equilibrium?
6. Kinetic-friction direction?
7. What provides centripetal force?
8. Maximum static friction?

Section B 2-mark questions - 6 x 2 = 12 marks

9. kg under net force 20 N: a?
10. 0.2 kg changes velocity +10 to -5 m/s: Deltap?
11. kg resting on level floor, g=9.8: N?
12. kg at 6 m/s in radius 3 m: force?
13. N for .2 s: impulse?
14. Why does swimmer push water backwards?

Section C 3-mark questions - 4 x 3 = 12 marks

15. kg trolley pulled 30 N, friction 6 N: a?
16. 0.15 kg ball at 20 m/s stops in .05 s: average force?
17. kg block on frictionless 30 deg incline, g=10: a,N?
18. kg at 5 m/s in radius 2.5 m: radial force?

Section D 4-mark questions - 2 x 4 = 8 marks

19. kg block released on frictionless 30 deg incline: speed after 3 s, g=10?
20. 0.5 kg ball rebounds +12 to -8 m/s in .1 s: average force?